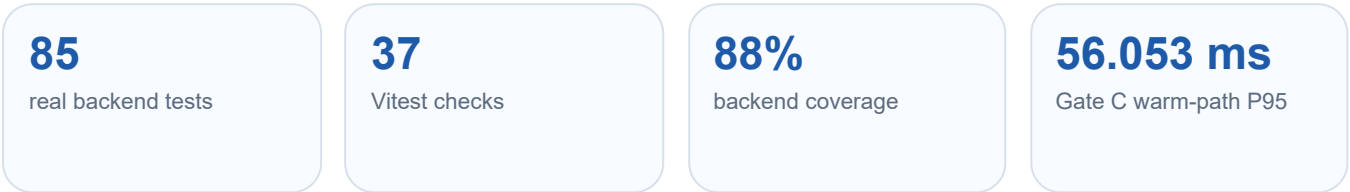


Shared care notes that make trust visible.

A clinic-scoped longitudinal collaboration layer for the moment when a care team needs to understand what changed and what needs action in under ten seconds.

Phase 9 final release candidate · optional DeepSeek · bounded prerecorded Voice fixture · FastAPI + React/TypeScript · Render PostgreSQL evaluation

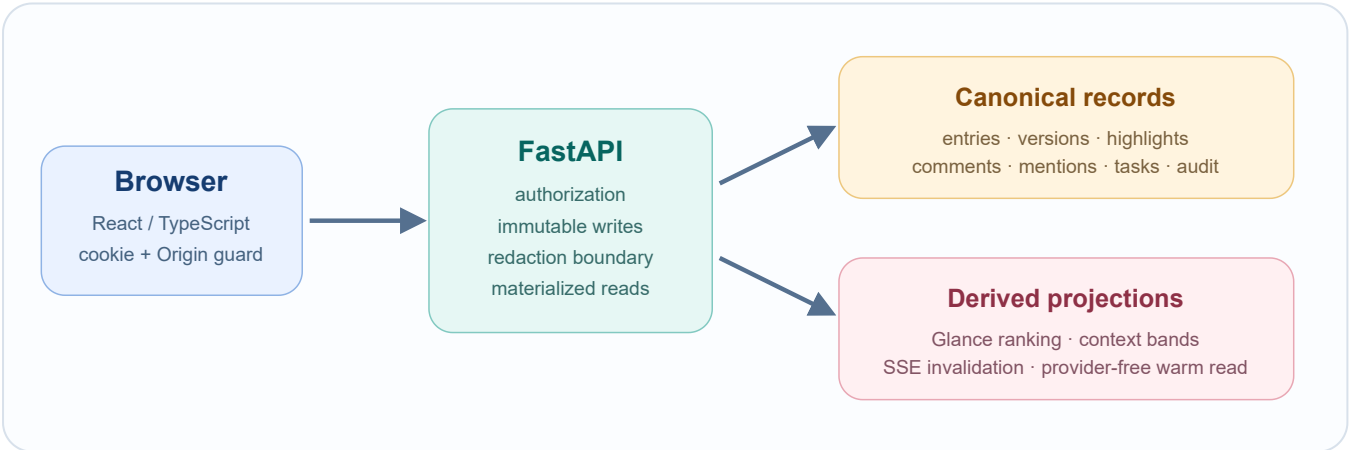


SYSTEM BOUNDARY

Product thesis

Nightingale is not an EHR replacement, diagnostic system, or autonomous medical decision-maker. AI-scribed content remains a suggestion until clinician review. Display ranking, explicit medical risk, provenance, and human confirmation are separate fields.

ARCHITECTURE



Server is canonical

Clinic scope and role permissions are enforced on every protected request. No UI-only role switch or database credential reaches the browser.

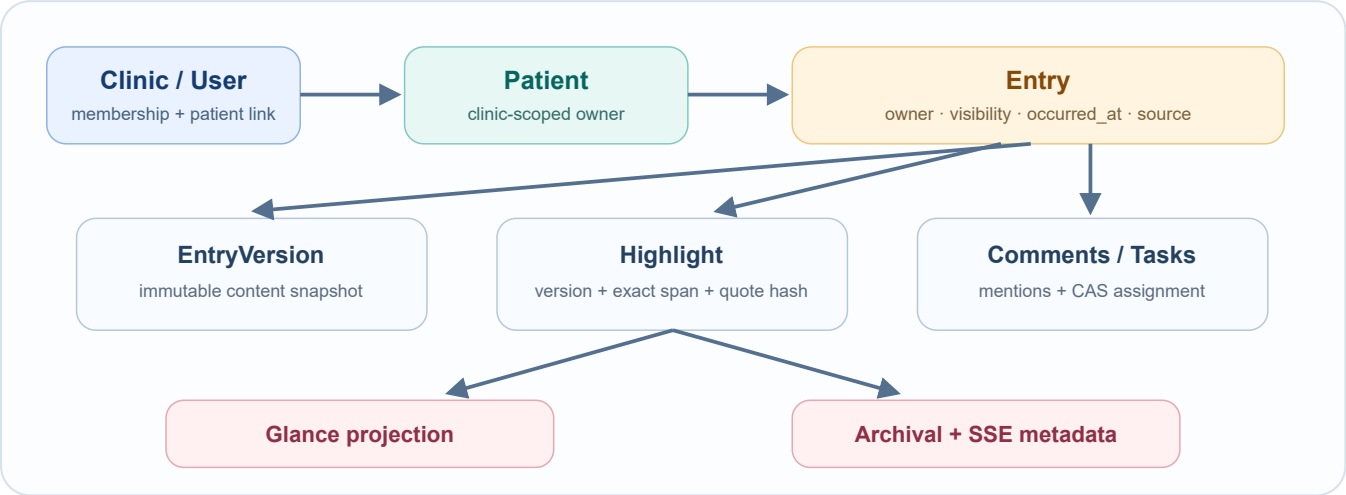
AI and Voice are bounded

Fixture remains the default. DeepSeek is optional and redaction-gated. Voice is prerecorded synthetic audio with a prepared timestamped transcript; no ASR inference is claimed. Glance reads never call a provider.

Canonical evidence stays intact while views stay glanceable.

The design separates stable record identity, immutable snapshots, source-linked highlights, human review, and derivative read models. Summaries are never the source of truth.

SCHEMA RELATIONSHIPS



EXACT PROVENANCE

Source navigation contract

`source\_entry\_id` + `source\_version\_id` + Python Unicode-codepoint offsets + exact quote + SHA-256. The frontend validates `Array.from(text)` and shows an integrity warning instead of an approximate match.

TRUST CONTROLS

Human decisions remain explicit

Suggested, accepted, rejected, superseded, and conflict-review states remain visible. A clinician can confirm semantic truth, but cannot erase the source record.

HYBRID RETRIEVAL BANDS

BAND	POLICY	READ SHAPE
Hot	Last 14 days, or protected by action/risk/conflict/discussion/pin/confirmation/care plan.	Full current detail.
Warm	14–90 days and not protected.	Metadata index; canonical source remains available.
Cold	Older than 90 days and not protected.	Derived period summary, manifest hash, and immutable source pointers.

```
final display priority = clamp(base + recency + explicit risk + open action + clinician confirmation + adaptive feedback, 0, 100)
```

Adaptive feedback is clinic-scoped, idempotent, structured-feature-only, and bounded to [-12, +12]. It changes display priority, never risk or provenance.

A reproducible local prototype with an honest production boundary.

Evidence below is synthetic and local. It establishes application behavior, not hosted compliance or PostgreSQL production guarantees.

QUALITY GATES

Backend

85 passed · 88% coverage under the reproducible command · Ruff · mypy · pip check. Alembic `0010\_postgres\_compat` passes fresh, legacy repair, downgrade/re-upgrade, and `alembic check` paths.

Frontend

37 Vitest tests · ESLint · Prettier · TypeScript · Vite build. Bilingual chrome, Guide focus, contextual drawers, exact text flow, preview dimensions, AI Scribe states, Voice fixture states, mentions, and reduced motion are covered.

Browser

18 Playwright checks at 1440×900 and 390×844: 14 core Gate B checks plus four isolated Voice fixture checks for scope, audio, timestamps, source navigation, Chinese chrome, mentions/tasks, SSE, revisions, 409 conflict, and patient privacy.

Warm path

P95 56.053 ms from 1,000 real TCP requests, concurrency 10, 26 patients, 208 materialized rows, zero errors. Local SQLite/Uvicorn approximation.

DEMONSTRATION PATH

A

Trace and trust

Staff opens Glance, follows an AI-scribed item to its exact immutable source, and keeps suggestion review separate from source content.

B

Collaborate without erasure

Staff edits a role-owned note, selects an @mention, creates an assigned task, and Clinician moves it through its lifecycle while reviewing a plan revision.

C

Surface conflict and history

Two writes use one expected version; the stale write returns **409** and remains beside the winner. Historical context discloses a derived summary that is not the original record and offers labelled immutable original-record rows.

TRADE-OFFS AND DEFERRED WORK

What is implemented: server-side RBAC and clinic scope, immutable revisions, exact provenance, comments/mentions/tasks, metadata-only SSE invalidation, conflict preservation, redaction-gated deterministic AI jobs, prerecorded synthetic audio with prepared timestamped transcript fixtures, bounded adaptive ranking, derived archival context, materialized warm reads, and Render-ready Docker/Blueprint files.

What is not claimed: functional Whisper/ASR or model quality, DeepSeek provider compliance, clinical production readiness, production retention policy, microphone/diarization, or final video content QA before human review.

Safety sentence

AI output is a suggestion. Derived summaries are not canonical sources. Risk is not learned silently. Human review is not replaced by ranking.